



To: Dr. Steven Mitchell
Topic: HW6 Grade
From: Jaegon Hibbits

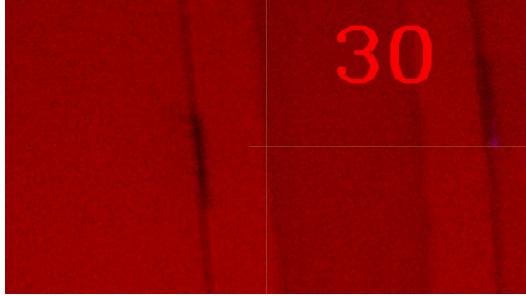
I believe my lane detection implementation for HW6 was marked down inconsiderably. This paper will argue why the homework submission proves more complete than the original grade.

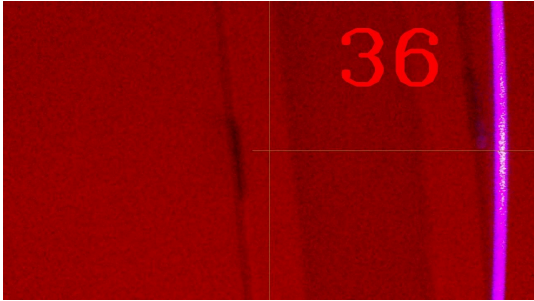
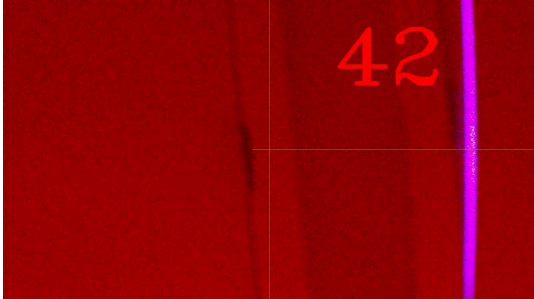
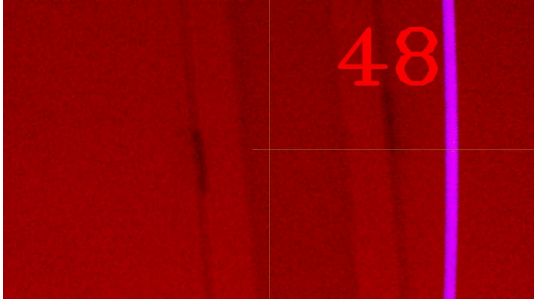
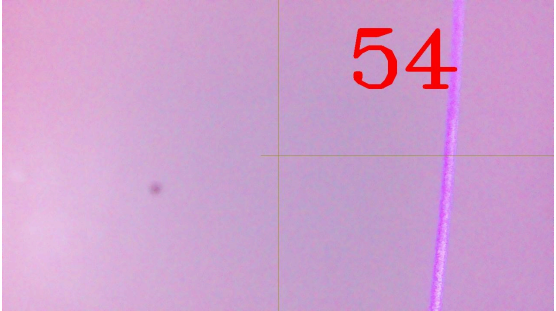
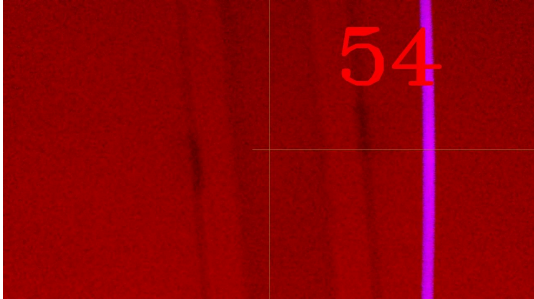
The only necessary **code** changes are in correspondence to shifting data output (graph) from referencing the center of the image to the edge of the image. As I was referencing the center, my graph was significantly different, almost a mirror of the example provided in the assignment. The more pressing issue was the fact not enough of my data points were being collected, this was purely due to bad lighting, as in a dark setting (new data collection of images), the output is ideal.

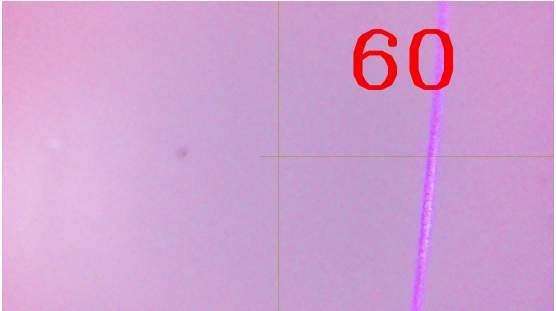
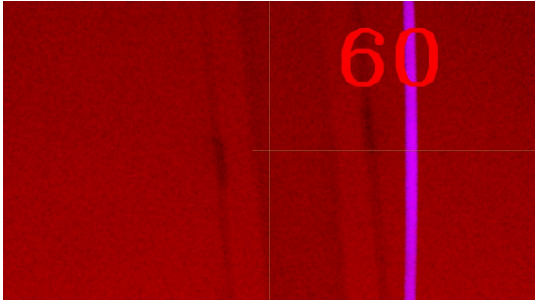
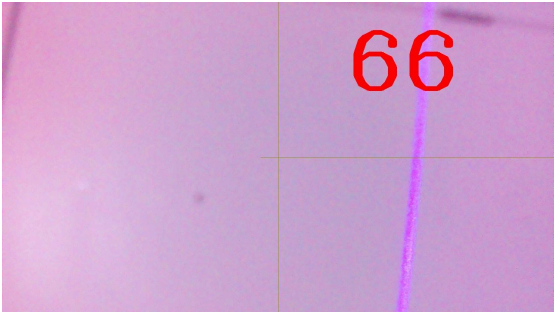
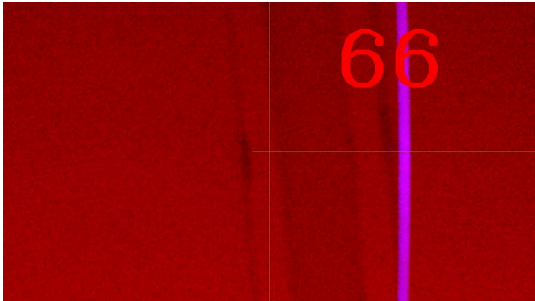
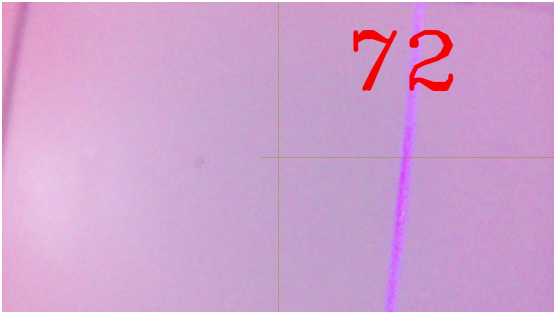
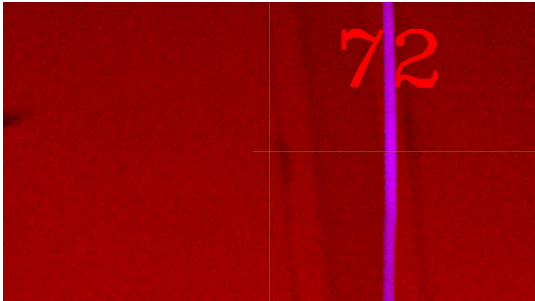
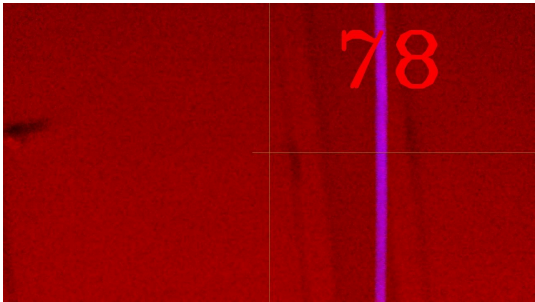
These were my specific code changes:

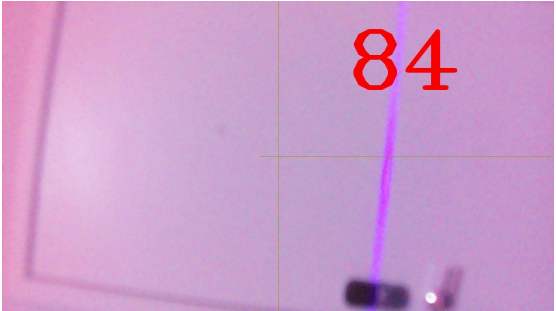
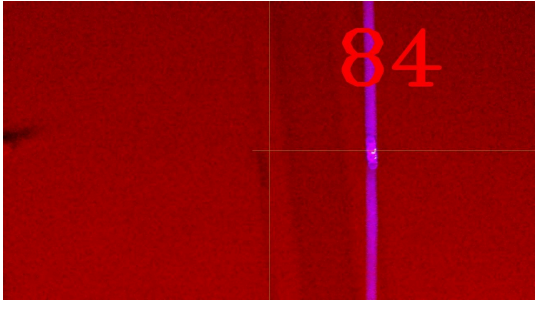
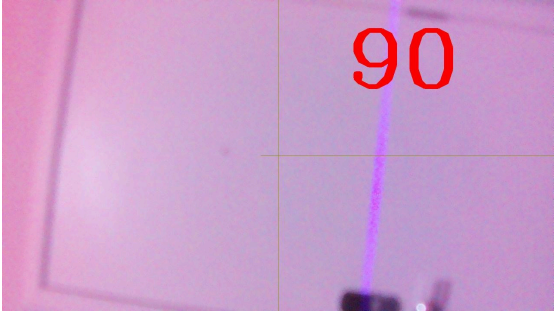
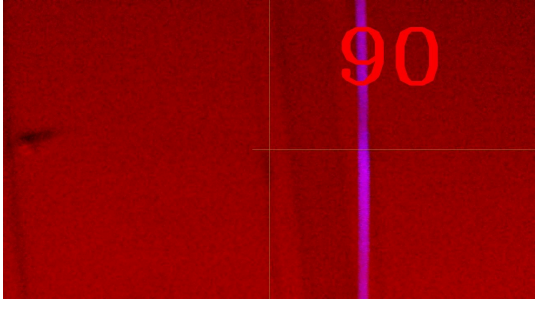
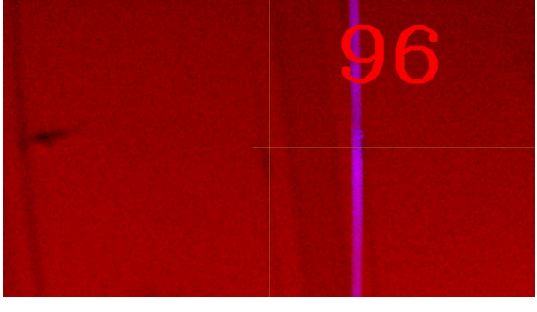
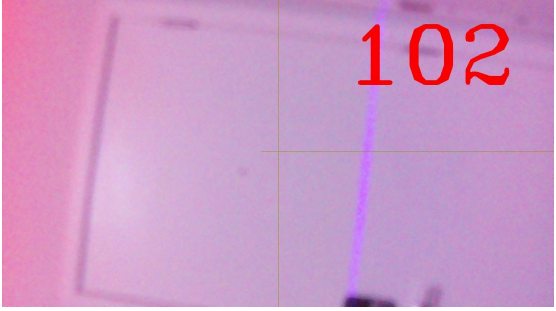
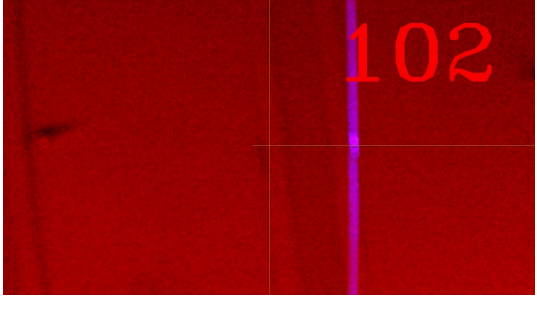
- mirror the pixel using $1280 - \text{raw_pixel}$ so graph matches assignment reqs
 - `image_width = 1280`
 - `calibration_pixel = image_width - raw_pixel`
 - `x_plot.append(calibration_pixel)`
 - `plt.xlabel("Pixel")`
- Ro and RpC changes (calibration)
 - `Ro = 0.41`
 - `Rpc = -0.00068`

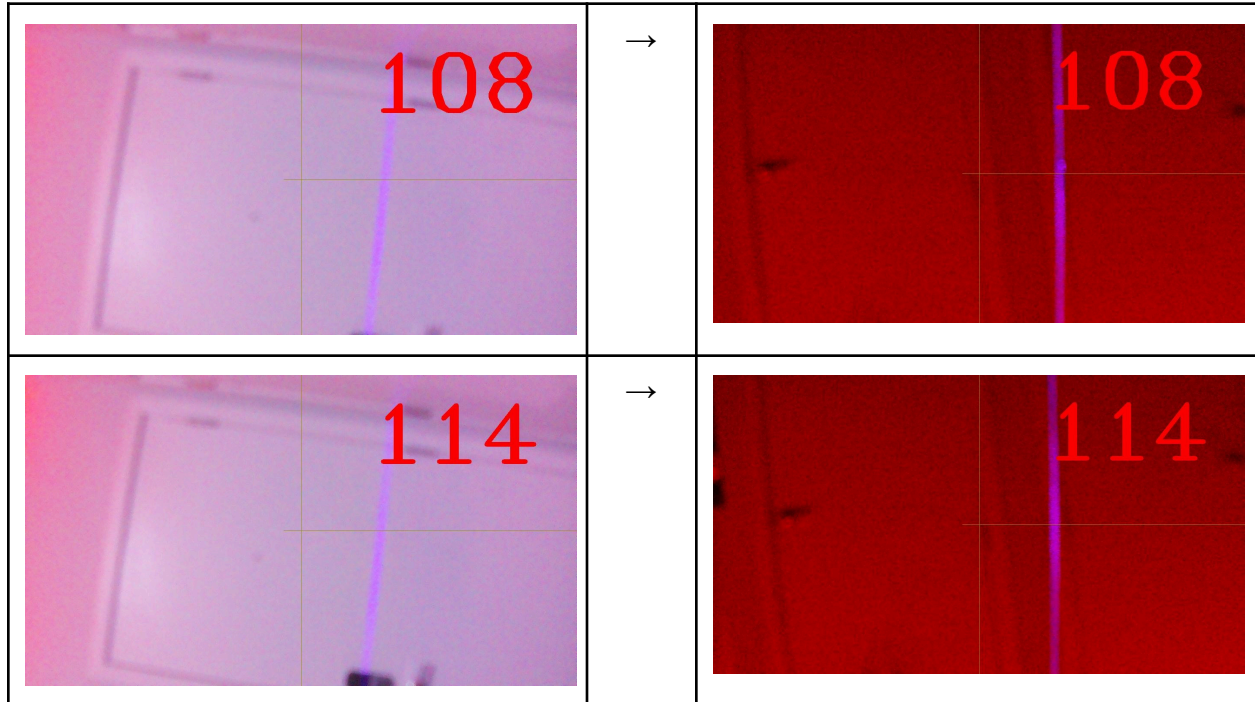
Image changes:

Original	→	Improved
	→	

 A micrograph showing a purple background with a small dark spot in the upper left and a vertical purple line on the right. The number "36" is in red in the top right.	→	 A micrograph showing a dark red background with a vertical purple line on the right. The number "36" is in red in the top right.
 A micrograph showing a purple background with a small dark spot in the upper left and a vertical purple line on the right. The number "42" is in red in the top right.	→	 A micrograph showing a dark red background with a vertical purple line on the right. The number "42" is in red in the top right.
 A micrograph showing a purple background with a small dark spot in the upper left and a vertical purple line on the right. The number "48" is in red in the top right.	→	 A micrograph showing a dark red background with a vertical purple line on the right. The number "48" is in red in the top right.
 A micrograph showing a purple background with a small dark spot in the upper left and a vertical purple line on the right. The number "54" is in red in the top right.	→	 A micrograph showing a dark red background with a vertical purple line on the right. The number "54" is in red in the top right.

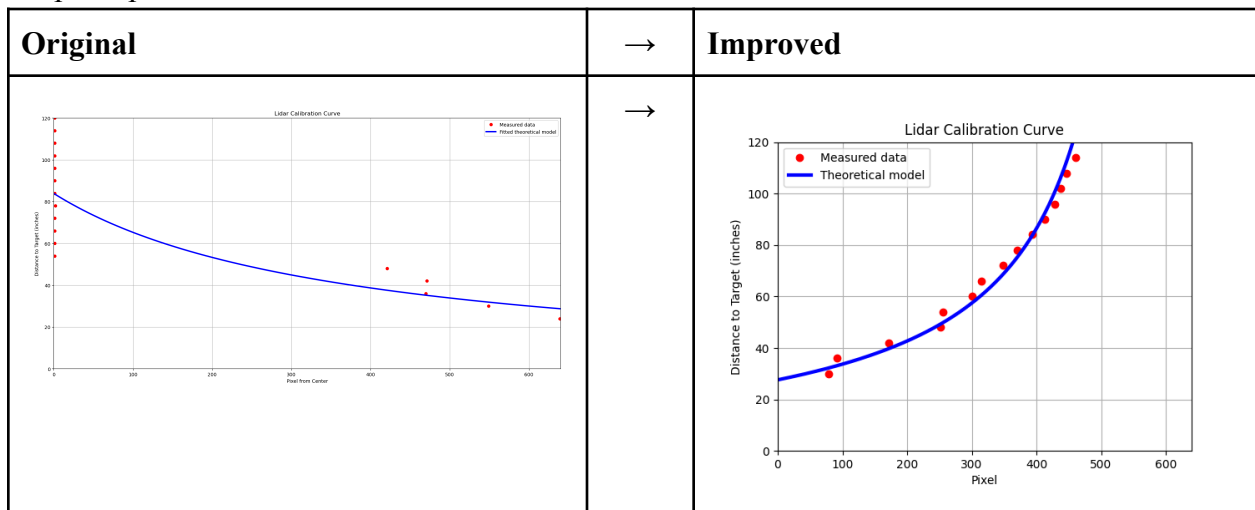
 A light micrograph showing a tissue section stained with hematoxylin and eosin (H&E). The tissue appears pinkish-purple. A red number "60" is overlaid in the upper right quadrant.	→	 A dark-field micrograph of the same tissue section as the left image. The tissue appears dark red. A red number "60" is overlaid in the upper right quadrant.
 A light micrograph showing a tissue section stained with hematoxylin and eosin (H&E). The tissue appears pinkish-purple. A red number "66" is overlaid in the upper right quadrant.	→	 A dark-field micrograph of the same tissue section as the left image. The tissue appears dark red. A red number "66" is overlaid in the upper right quadrant.
 A light micrograph showing a tissue section stained with hematoxylin and eosin (H&E). The tissue appears pinkish-purple. A red number "72" is overlaid in the upper right quadrant.	→	 A dark-field micrograph of the same tissue section as the left image. The tissue appears dark red. A red number "72" is overlaid in the upper right quadrant.
 A light micrograph showing a tissue section stained with hematoxylin and eosin (H&E). The tissue appears pinkish-purple. A red number "78" is overlaid in the upper right quadrant.	→	 A dark-field micrograph of the same tissue section as the left image. The tissue appears dark red. A red number "78" is overlaid in the upper right quadrant.

 84	→	 84
 90	→	 90
 96	→	 96
 102	→	 102



Well clearly it should also be noted that the re-calibration had significant affect of vertical laser-camera improvements as well.

Graph improvements



Request:

If possible, I am requesting credit for the HW; I received a zero with a working script, purely an error due to human incompetence and failure to find ideal parameters. Thank you.